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PROFESSIONAL SUMMARY

Systems Engineer with over 4 years of experience as a **Full Stack Developer** and **Data Engineer**, specializing in scalable web application development with **Angular** and **Laravel**, as well as building **ETL pipelines** for processing and analyzing large volumes of data. With a strong focus on **Software Architecture**, **Clean Code**, and **agile methodologies (Scrum)**, I have led high-performance system optimization and implemented comprehensive data solutions.

My experience includes developing **RESTful APIs** with **Laravel**, **Node.js**, and **Python**, version migrations of **Laravel** and **PWA to Tauri.js**, cloud solutions implementation with **AWS (EC2, RDS, Lambda)** and **Supabase**, queue management with **Laravel Queues**, and process automation with **AWS Lambda**. I possess solid knowledge in **Python (pandas, SQLAlchemy)**, **ETL**, **dimensional data modeling (star schema)**, and **SQL/NoSQL databases**, applied in analytics and business intelligence projects.

PROFESSIONAL EXPERIENCE

Full Stack Developer

Feb 2026 - Present

Helpfull Qualitative Research

- Led the migration from **Laravel 7.4 to 8.2**, updating dependencies, refactoring legacy code, and ensuring compatibility with new framework features.
- Migrated a **PWA (Progressive Web App)** to a desktop application using **Tauri.js**, generating native executables for Windows, macOS, and Linux, expanding the platform's reach to desktop environments.
- Implemented a **whitelist** system for access control based on authorized IPs and domains, enhancing application security.
- Managed processing queues with **Laravel Queues** for asynchronous tasks such as email sending, report generation, and background data processing.
- Configured the production environment on **AWS EC2**, deploying the Laravel application with **Nginx** and **PM2** for process management.
- Implemented databases on **AWS RDS (PostgreSQL)** and **Supabase**, configuring **Row Level Security (RLS)** policies for data access control.
- Developed **automatic PDF download** functionality using **Dompdf** and **Laravel PDF**, generating on-demand custom reports.
- Designed **semi-automatic templates** with custom **Artisan commands**, enabling dynamic content generation based on user requests.
- Developed responsive interfaces using **Laravel Blade**, **CSS**, and **TailwindCSS**, improving user experience and load speed.

Full Stack Developer

Feb 2024 - Feb 2026

Constructora Ariguani S.A.S

- Led the optimization of a web application used by over 700 internal users, reducing load times by 50% through ORM query refactoring and *caching* strategy implementation.
- Redesigned the system architecture under the MVC pattern using **Laravel** on the backend and **Angular** on the frontend, improving scalability and code maintainability.
- Developed **RESTful APIs** in **Laravel** and **Python** for data integration between systems, ensuring proper database normalization (3NF) and index creation for optimal performance.
- Implemented functional modules in **Angular** with **RxJS** and **TypeScript**, creating reusable components and enhancing user experience.
- **Developed the complete Daily Report module** from scratch, enabling registration of worked hours and interruptions for both company employees and subcontractors, optimizing attendance and productivity control.
- Implemented significant improvements in the **Equipment module**, optimizing machinery and tool assignment and tracking on construction sites.
- Made enhancements to the **Materials module**, streamlining registration, tracking, and inventory control of construction materials.
- Optimized the **Concrete Request module**, improving response times and integration with the production workflow.
- Implemented an executive **dashboard system in Power BI**, connecting multiple data sources to facilitate management decision-making.
- Improved the performance and functionality of the **Material Request dashboard**, reducing load times and enhancing visualization of critical data.
- Optimized the **Releases dashboard**, improving load speed and accuracy of construction progress reports.

Full Stack Developer

Dec 2023 - Feb 2024

Helpfull Qualitative Research

- Maintained stability of a platform with 1,000 active users, achieving 90% availability through proactive correction of critical backend errors with **Laravel** and **PHP**.
- Managed adjustment and customization of dynamic templates in **Laravel Blade**, adapting the interface to changing client requirements.
- Performed migrations and query optimizations in **MySQL**, ensuring optimal database performance.

Full Stack Developer

May 2023 - Jan 2024

Brace Developers

- Prepared comprehensive technical documentation of the application, detailing architecture, API endpoints, and data model, facilitating onboarding of new developers.
- Automated critical financial processes with **AWS Lambda**, reducing execution time of repetitive tasks by 85% and eliminating manual errors in credit notes.
- Implemented new functional modules using **Angular** on the frontend and **NestJS** on the backend, following **SOLID** principles and development best practices.
- Developed reusable components in **Angular** with **RxJS** and **TypeScript**, improving code maintainability.

Full Stack Developer

Sep 2022 - Feb 2023

DevZeros

- Designed and implemented a report export system in Excel for the ICBF's ORFEO system, reducing report generation time by 70%.
- Developed dynamic templates in **CodeIgniter (PHP)** for custom document generation.
- Performed maintenance tasks and documentation generation for the SIG Pacifico Colombia platform, ensuring service continuity.

Data Engineer / ETL Developer

Feb 2026 - Apr 2026

Graduation Project - Universidad Popular del Cesar

- Designed and built a complete **ETL pipeline** in Python for processing Colombian higher education data from **SNIES** (enrollment, registered, admitted, graduates) and **SPADIES** (dropout) sources.
- Implemented a modular architecture with 5 stages: **Extract, Transform, Validate, Load, and Control (audit)**.
- Developed extraction modules with **pandas** to read Excel (xlsx, xlsb) and CSV files from multiple sources, including dynamic column detection by content.
- Implemented advanced transformations including Unicode normalization (NFKD), data cleaning, filtering by specific institutions (codes 1120 and 1123), and mapping to a canonical schema.
- Designed a **star schema data model** with dimension tables and fact tables for SNIES and SPADIES.
- Developed idempotent load modules using **SQLAlchemy** and **PostgreSQL**, implementing UPSERT strategies to avoid duplicates.
- Implemented an **audit system** with a control table (etl_control) to log each load and ensure process traceability.
- Containerized the solution with **Docker** and **Docker Compose**, enabling reproducible deployment in any environment.
- Optimized pipeline performance processing over **10,000 records per second** with bulk loading and deduplication strategies.

EDUCATION AND CERTIFICATIONS

Systems Engineering

2017 - 2026

Universidad Popular del Cesar

Expected: July 3, 2026

Certifications

- **Cloud Architecture Basic Level** - Talento Tech 2025
Official certification with completion certificate.
- **Introduction to Data Science** - Cisco 2024

TECHNICAL SKILLS

Languages and Frameworks

JavaScript (ES6+), TypeScript, Angular, React.js, Laravel (PHP), Node.js (NestJS), Python, Java, CodeIgniter, TailwindCSS

Data Engineering & ETL

Python (pandas, SQLAlchemy), ETL Pipelines, Dimensional Modeling (Star Schema), Data Auditing, Unicode Normalization

Databases

PostgreSQL, SQL Server, Oracle, MySQL, MongoDB, Supabase (RLS)

Cloud & DevOps

AWS (EC2, RDS, Lambda, S3), Docker, Docker Compose, CI/CD, Linux (Bash), Nginx, PM2

Tools and Technologies

Tauri.js, Laravel Queues, Laravel Artisan, Dompdf, RxJS, Git, SonarQube, Power BI

Methodologies

Scrum, Clean Code, Design Patterns, SOLID, Agile Methodologies

Note: Agile methodologies (Scrum, XP) applied in all roles, managing sprints, retrospectives, and iterative value delivery.